

Network Topologies (Physical & Logical)

◆ What Are Network Topologies?

A **network topology** describes *how computers, devices, switches, and routers are arranged* and how data flows among them.

There are **two perspectives**:

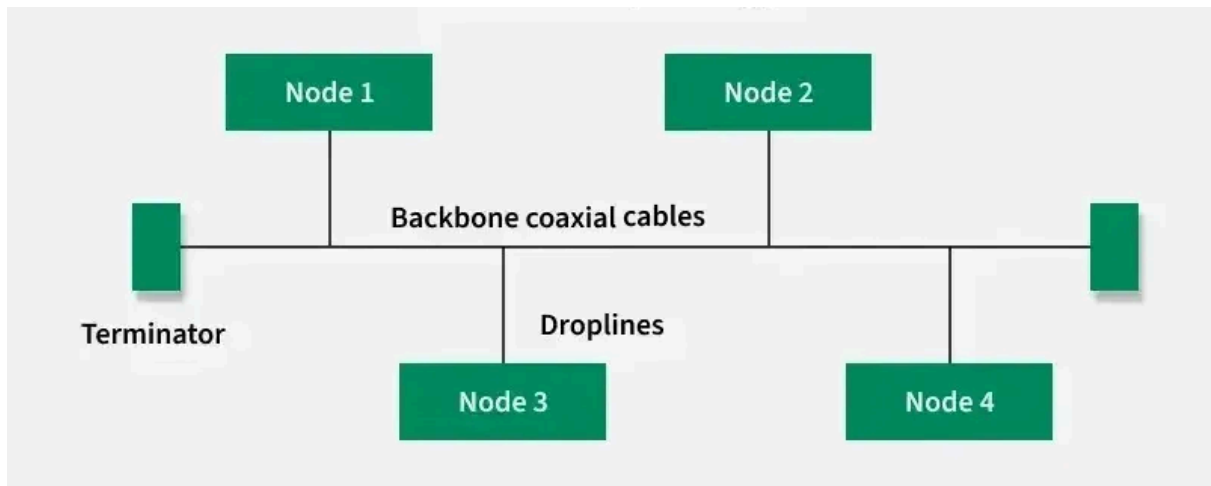
- **Physical topology:** Actual layout of cables & devices
- **Logical topology:** How data flows, regardless of physical wiring

Choosing the right topology affects:

- ✓ Network performance
- ✓ Scalability
- ✓ Fault tolerance
- ✓ Cost
- ✓ Security
- ✓ Maintenance complexity

Let's break down each one **in depth** 📌

◆ Bus Topology — The Simplest Backbone-Based Network



How it works:

All devices share one central communication cable called a *bus*.

Advantages:

- ✓ Very cheap to set up
- ✓ Uses fewer cables
- ✓ Best for small networks

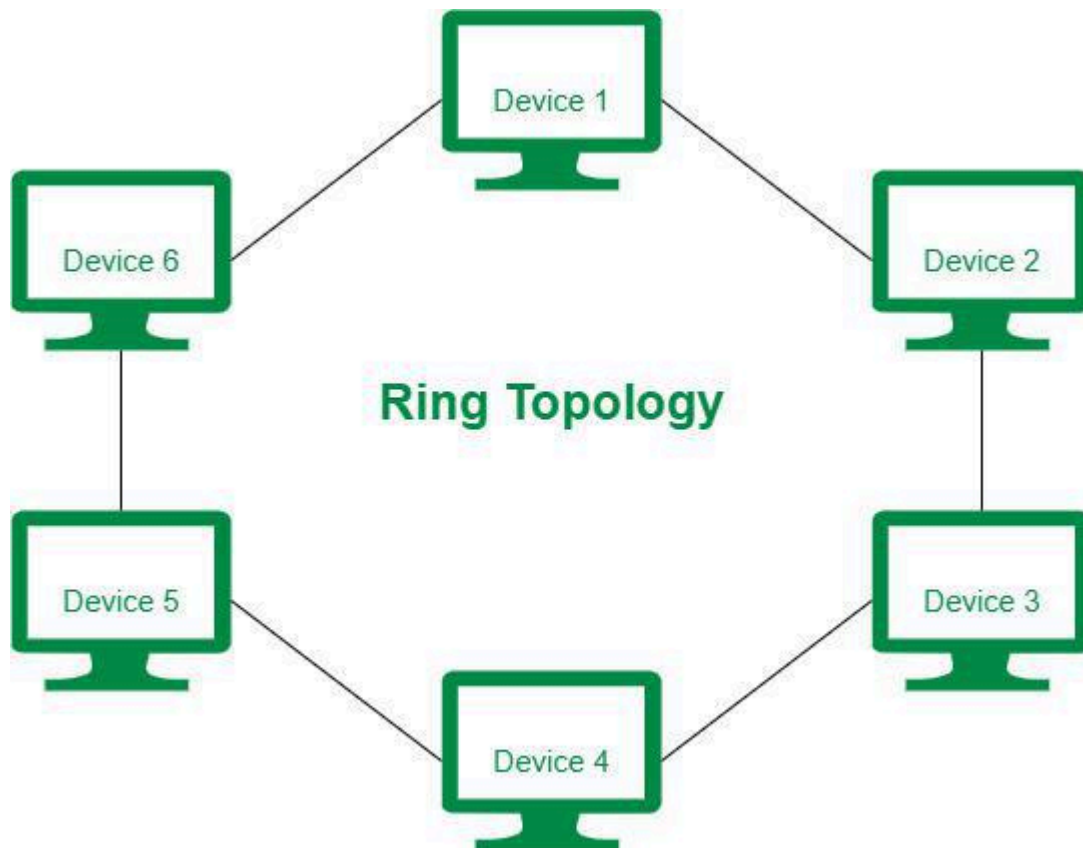
Disadvantages:

- ✗ Backbone cable failure = ENTIRE network fails
- ✗ Performance drops as more devices connect
- ✗ Difficult to troubleshoot
- ✗ Not secure — data is broadcasted to all devices

Used In:

- Small home networks (older systems)
- Temporary networks
- Early LANs (Ethernet 10BASE-2/10BASE-5)

◆ Ring Topology — Token Passing Model



How it works:

Devices are connected in a circle.

Data travels in one direction using a **token** — a special control frame.

Advantages:

- ✓ Predictable performance
- ✓ No collisions
- ✓ Equal access to the network

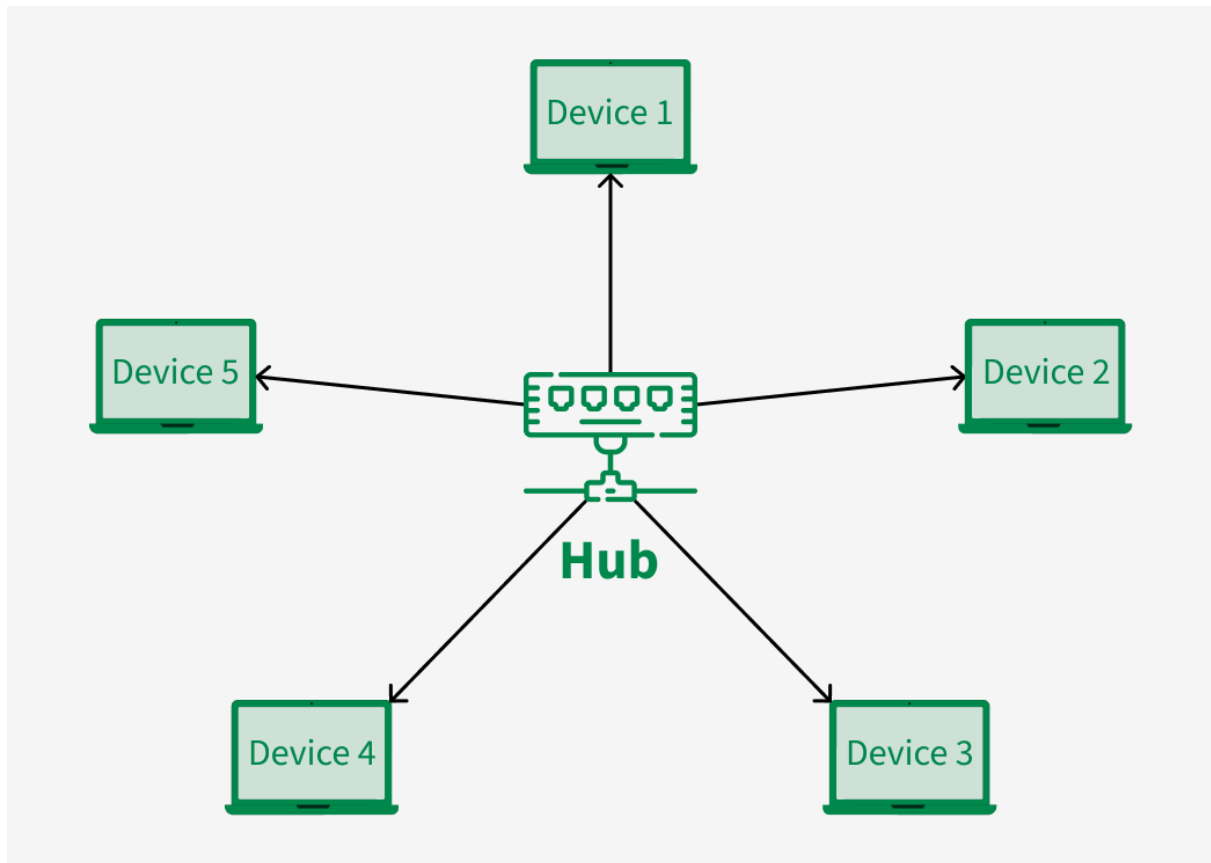
Disadvantages:

- ✗ One device failure = whole ring breaks
- ✗ Difficult to reconfigure
- ✗ Not scalable

Used In:

- Token Ring networks (IBM)
- FDDI (Fiber Distributed Data Interface)

◆ Star Topology — Most Common Modern LAN Topology



How it works:

Each device connects to a central switch/hub.

Advantages:

- ✓ Very easy to manage
- ✓ Failure of one device doesn't affect others
- ✓ Highly scalable
- ✓ Performance is high due to dedicated links
- ✓ More secure than bus or ring

Disadvantages:

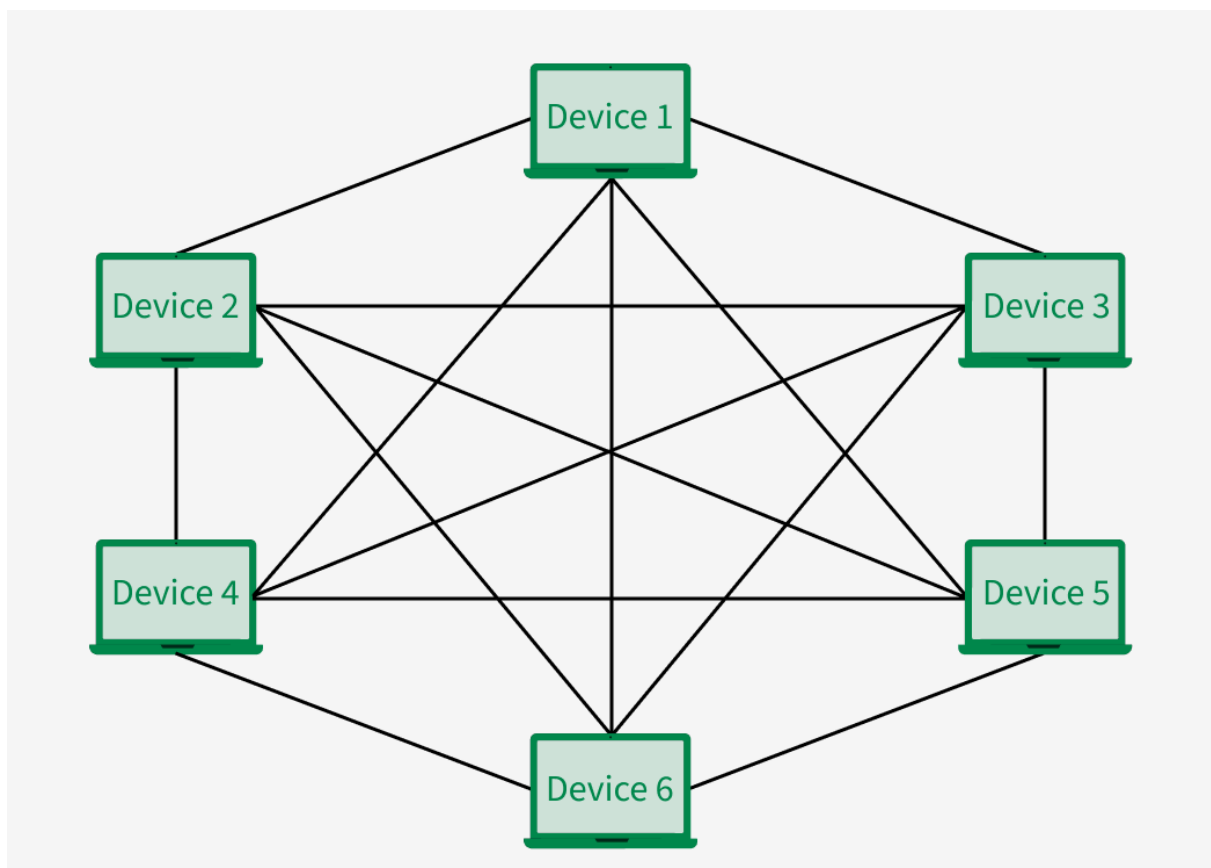
- ✗ Central device failure = network failure

✗ Requires more cable

Used In:

- Most modern Ethernet LANs
- Home Wi-Fi networks
- Corporate office networks

◆ Mesh Topology — Surgeons of Reliability & Redundancy



Two types:

- **Full Mesh:** Every node connected to every other
- **Partial Mesh:** Only critical nodes have multiple connections

Advantages:

- ✓ Maximum reliability

- ✓ No single point of failure
- ✓ Best performance
- ✓ Great for mission-critical networks

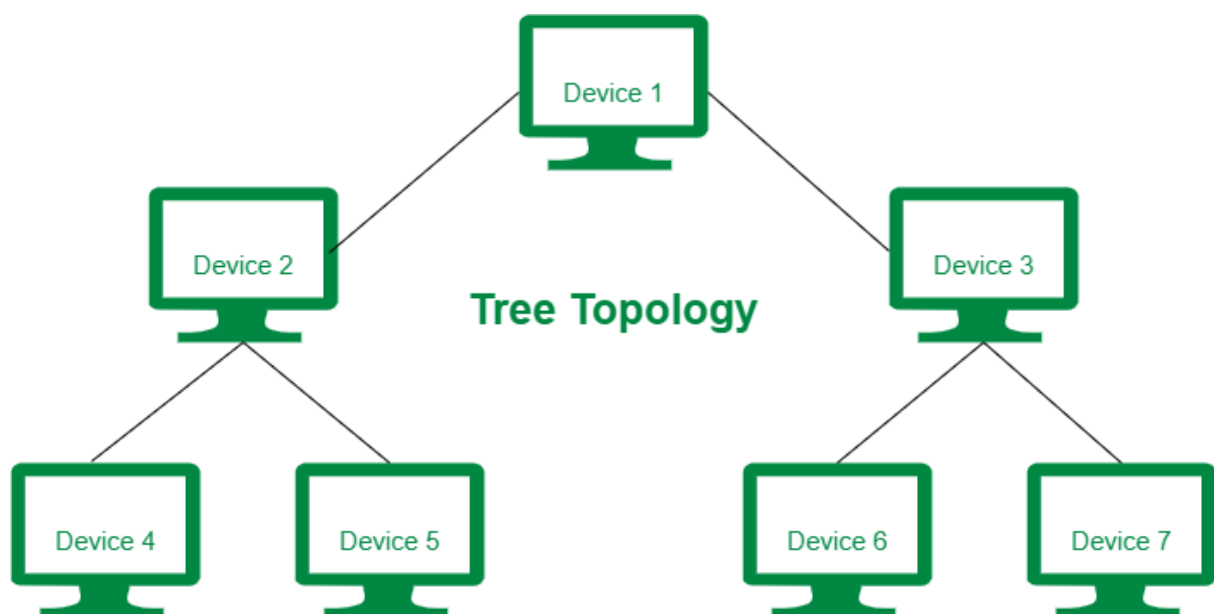
Disadvantages:

- ✗ Expensive (cables + NICs)
- ✗ Complex setup
- ✗ Hard to scale

Used In:

- Military networks
- Data centers
- Backbone of telecom networks
- Blockchain peer-to-peer systems

◆ **Tree Topology — Hierarchical Network Structure**



How it works:

A combination of star networks arranged in a hierarchy (parent-child model).

Advantages:

- ✓ Easy to expand
- ✓ Well-organized large networks
- ✓ Fault isolation is simple

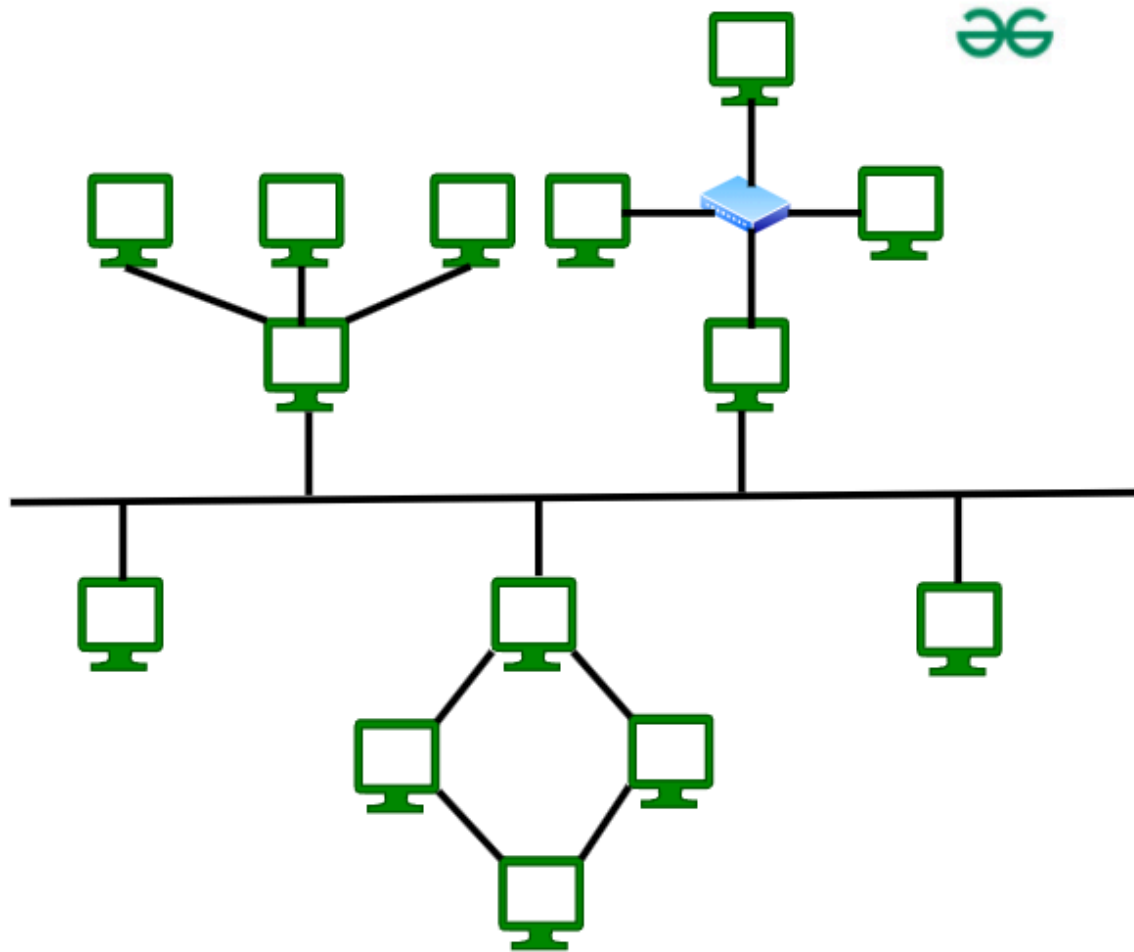
Disadvantages:

- ✗ High cabling requirements
- ✗ Root device failure = affects entire branch
- ✗ More complex than star

Used In:

- Corporate enterprise networks
 - ISP network architecture
 - University campus networks
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Hybrid Topology — Combination of Two or More Topologies



Examples:

- **Star-Bus** (most common)
- **Star-Ring**
- **Mesh-Star**

Advantages:

- ✓ Very flexible
- ✓ Can be designed to meet specific needs
- ✓ Scalable and efficient

Disadvantages:

- ✗ Expensive
- ✗ Complex troubleshooting

✗ Requires expert administration

Used In:

- Real-world modern networks
 - Enterprise + Cloud + Branch networks
 - Metropolitan area networks (MANs)
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◆ Logical vs Physical Topology (Often Confused)

Physical Topology:

How devices & cables are physically arranged

— "What the network *looks like*."

Logical Topology:

How data flows between nodes

— "How the network *behaves*."

Example:

A **physical star** network using Ethernet behaves **logically like a bus** (broadcast domain).

◆ Security Perspective: Which Topology Is Strongest?

Topology	Security Level	Notes
Bus	✗ Weak	Broadcast network — easy sniffing
Ring	✗ Weak	Single break = exposure
Star	★ Good	Central switch controls security
Tree	★★ Strong	Hierarchical ACL & segmentation
Mesh	★★★★ Very Strong	Hard to attack, highly redundant
Hybrid	★★★★ Strong	Depends on design

Mesh + VLAN segmentation + firewalls → Most secure.

◆ **Which Topology Should You Choose?**

Small Networks:

Star

Medium Businesses:

Star + Tree

Corporate / Enterprise:

Tree + Partial Mesh + VLANs

Data Centers / Telecom:

Full Mesh + Spine-Leaf topology

◆ **Final Summary**

Network topologies determine how devices connect, communicate, and stay resilient.

- **Bus & Ring** = legacy, rarely used
- **Star** = modern LAN standard
- **Mesh** = maximum reliability
- **Tree** = scalable corporate model
- **Hybrid** = real-world networks combining strengths of multiple designs

Understanding these models is essential for:

- ✓ Network engineers
 - ✓ Ethical hackers
 - ✓ Cybersecurity analysts
 - ✓ VAPT professionals
 - ✓ Students learning networking fundamentals
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